Three tabulated-μ operators side-by-side: Chebyshev strict-h=0 vs Linear-CDF kernel vs Linear-CDF strict-h=0

Fixed-Point Factory — γ-sweep, same architecture three ways

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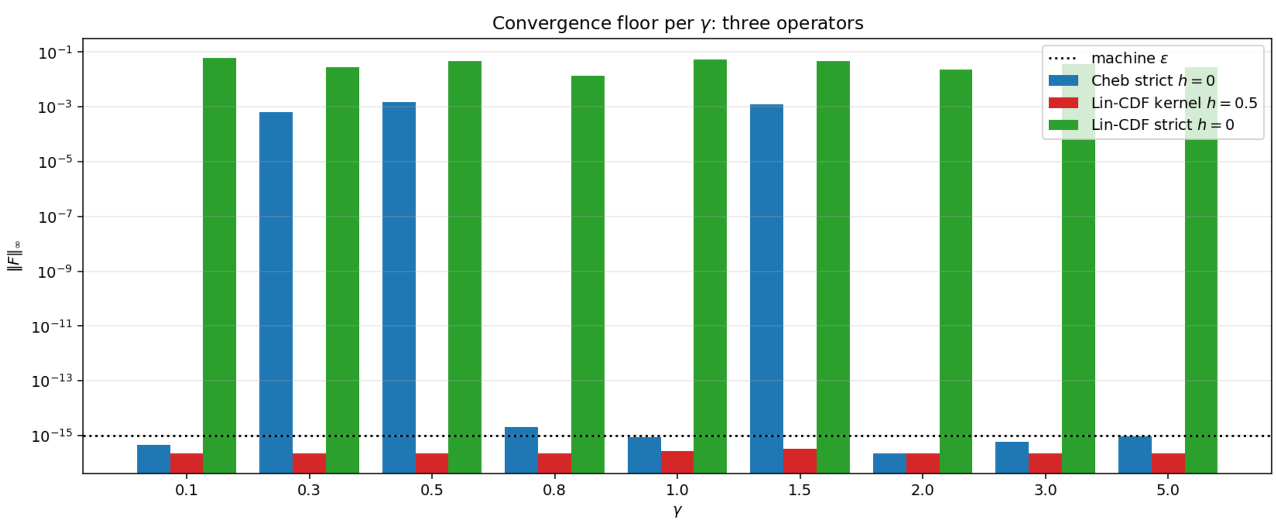
The three operators

All three share the tabulated-μ architecture (build $\mu(p, u_k)$ table, look up per cube cell, run CRRA clearing). They differ in:

	Cheb strict h=0	**Lin-CDF kernel**	**Lin-CDF strict h=0**
u-grid	Lobatto-atanh	Gaussian quantiles	Gaussian quantiles
P representation	tensor Cheb poly	piecewise bilinear	piecewise bilinear
contour integral	strict h=0 line int.	Gaussian band h=0.5	strict h=0 line int.
POU partition-of-unity	yes	— (kernel)	yes
ms / Φ (G=7)	540	6	7

The third operator (NEW): strict h=0 line integral, no kernel, just finds piecewise-linear root crossings on the CDF grid, weighted by partition-of-unity over both axes.

1 Convergence floor per γ



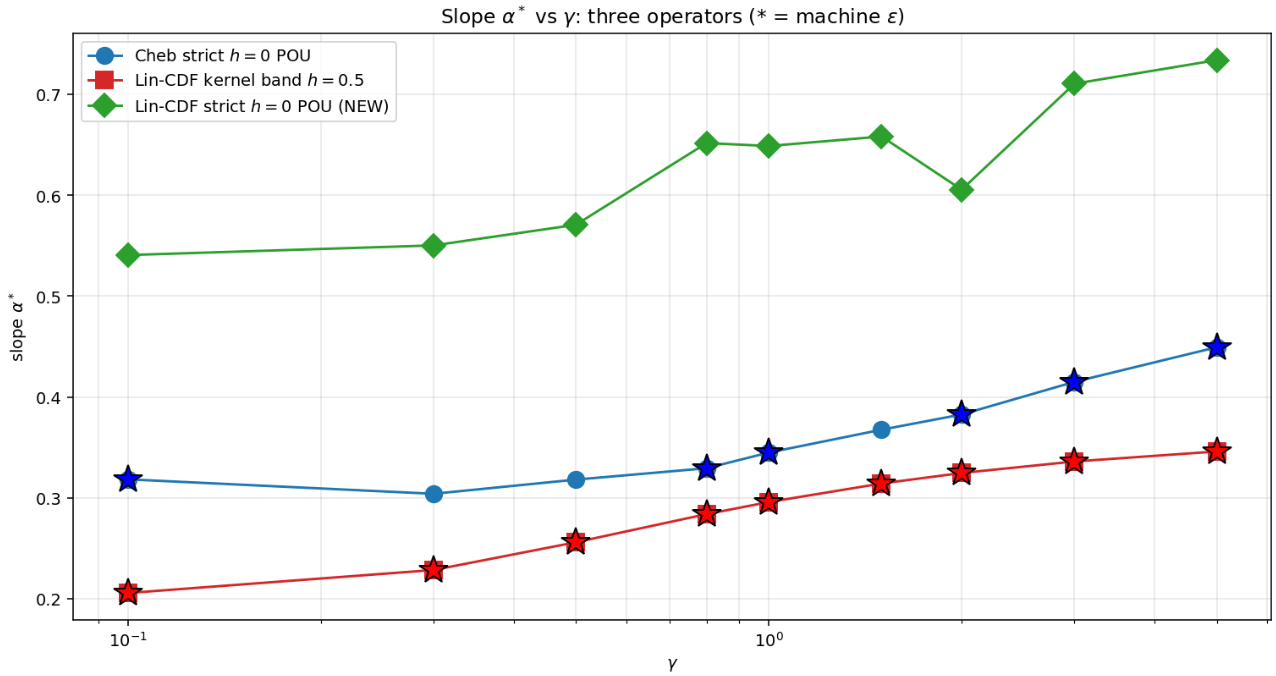
γ	Cheb strict $\|F\|_\infty$	Lin-CDF kernel $\|F\|_\infty$	Lin-CDF strict $\|F\|_\infty$
0.1	$4.4\times10^{-16}\star$	$2.2\times10^{-16}\star$	6.0×10^{-2}
0.3	6.4×10^{-4}	$2.2\times10^{-16}\star$	2.8×10^{-2}
0.5	1.5×10^{-3}	$2.2\times10^{-16}\star$	4.7×10^{-2}
0.8	$2.0\times10^{-15}\star$	$2.2\times10^{-16}\star$	1.4×10^{-2}
1.0	$8.9\times10^{-16}\star$	$2.8\times10^{-16}\star$	5.2×10^{-2}
1.5	1.3×10^{-3}	$3.3\times10^{-16}\star$	4.8×10^{-2}
2.0	$2.2\times10^{-16}\star$	$2.2\times10^{-16}\star$	2.3×10^{-2}
3.0	$5.8\times10^{-16}\star$	$2.2\times10^{-16}\star$	3.7×10^{-2}
5.0	$9.2\times10^{-16}\star$	$2.2\times10^{-16}\star$	2.9×10^{-2}

Verdict.

- **Lin-CDF kernel**: ALL 9 reach machine ε — fast, robust, but smoothed.
- **Cheb strict**: 6 of 9 reach machine ε — exact REE, but slow and N_Q sweet-spot-dependent.
- **Lin-CDF strict**: NONE reach machine ε — floors at $\sim 10^{-2}$. The piecewise-linear P has piecewise-constant $\partial P/\partial u$, which generates step discontinuities in the co-area integrand that POU cannot smooth out.

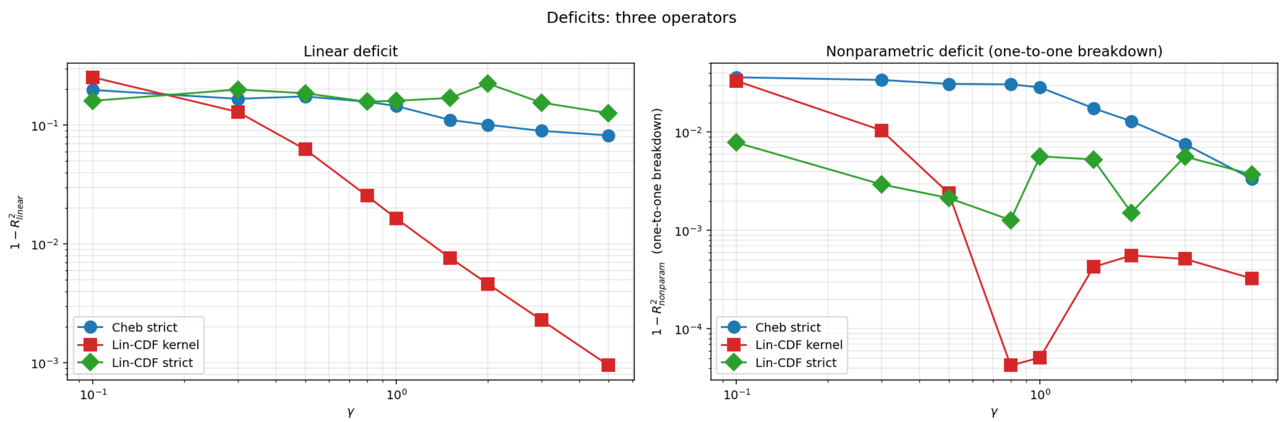
Why strict h=0 needs the Chebyshev representation. The co-area integrand $ff/|\nabla P|$ depends pointwise on the derivative ∇P. On the linear-CDF grid the derivative is piecewise constant and jumps at every grid line. As the root location $\xi_b^*(p)$ moves continuously with p, the value of $|\nabla P|$ at the root changes discontinuously each time the root crosses a grid line. POU smooths $|\nabla P| \to 0$ (turning points) but cannot smooth $|\nabla P|$ JUMPS. Chebyshev gives a C^∞ derivative everywhere, so no jumps.

2 FP slopes and deficits



The three operators give three different FPs:

γ	Cheb strict α^*	Lin-CDF kernel α^*	Lin-CDF strict α^*
0.1	0.318	0.205	0.541
0.3	0.304	0.228	0.550
0.5	0.318	0.256	0.571
0.8	0.329	0.284	0.652
1.0	0.345	0.296	0.649
1.5	0.368	0.314	0.658
2.0	0.383	0.324	0.606
3.0	0.415	0.336	0.711
5.0	0.449	0.346	0.734



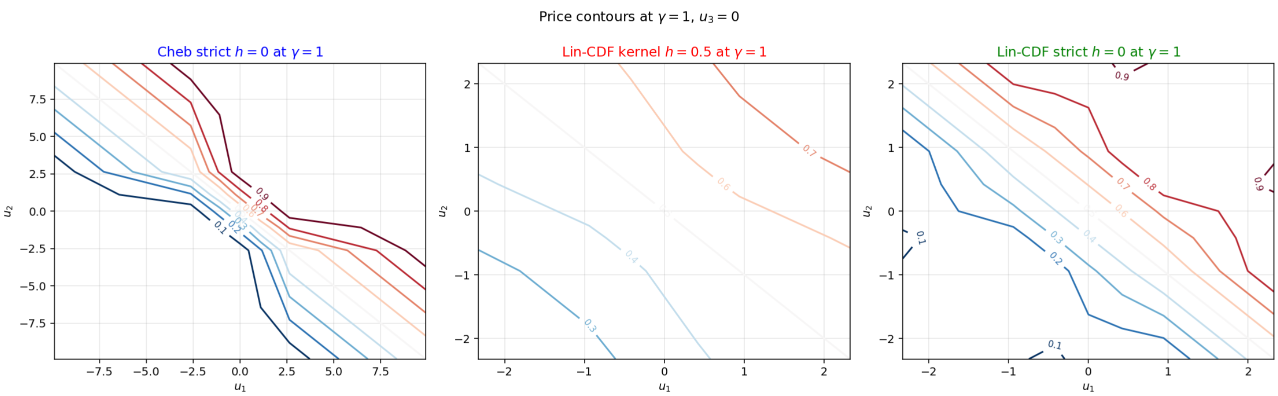
Pattern. For each γ:

- Lin-CDF kernel slope < Cheb strict slope < Lin-CDF strict slope
- The kernel smoothing biases the FP DOWN (less revealing).
- The strict-h=0-on-linear-grid biases UP (over-revealing) and never converges fully because of derivative jumps.
- Cheb strict is the only operator that's both exact AND converges to machine ε.

One-to-one breakdown.

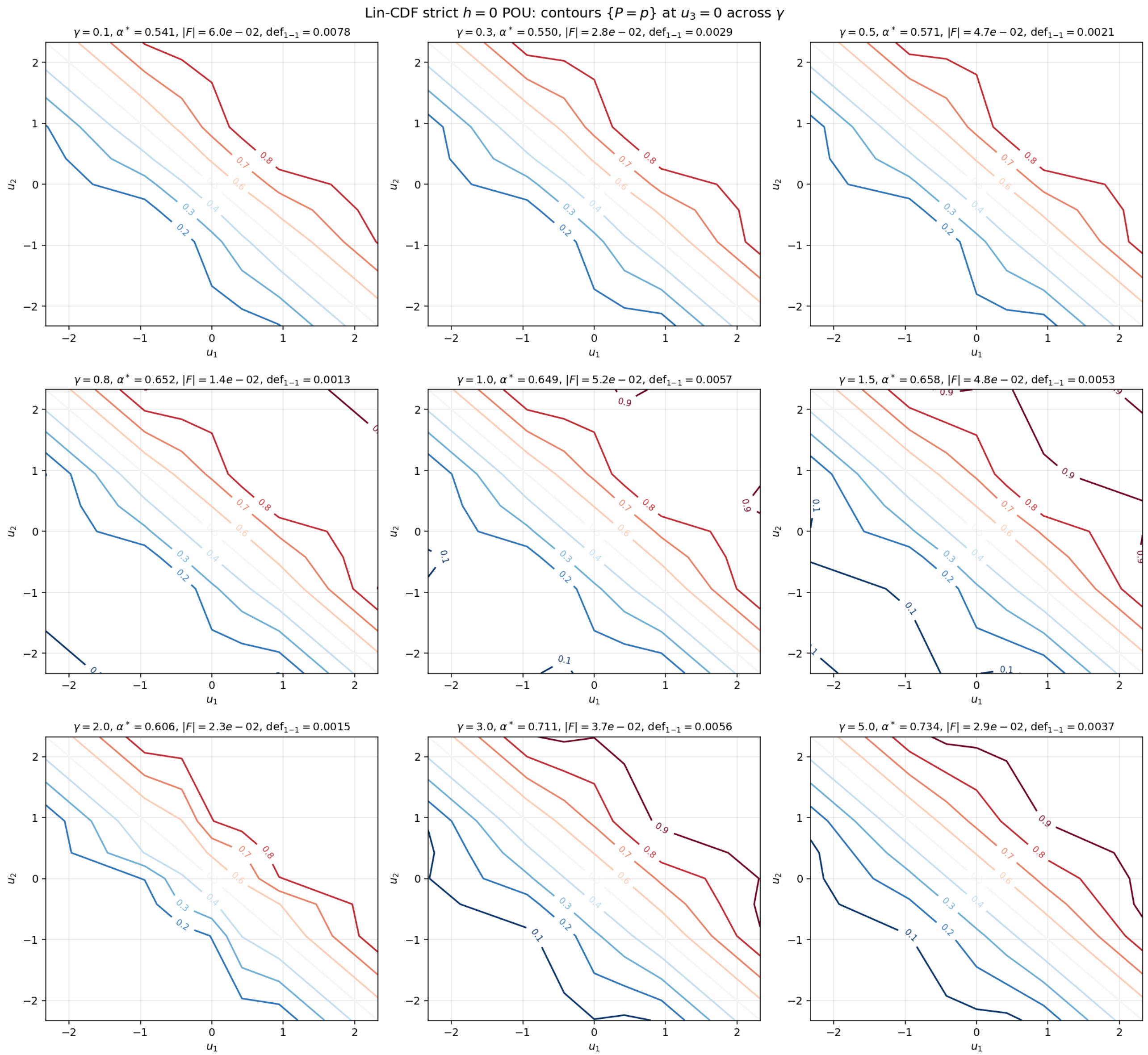
- Cheb strict: $0.003 - 0.036$ (genuine breakdown captured).
- Lin-CDF kernel: ~ 0 for $\gamma \geq 0.5$ (smoothed away by kernel band).
- Lin-CDF strict: $0.001 - 0.008$ (intermediate, partly reflective of the unconverged FP).

3 Contours at $\gamma = 1$, $u_3 = 0$

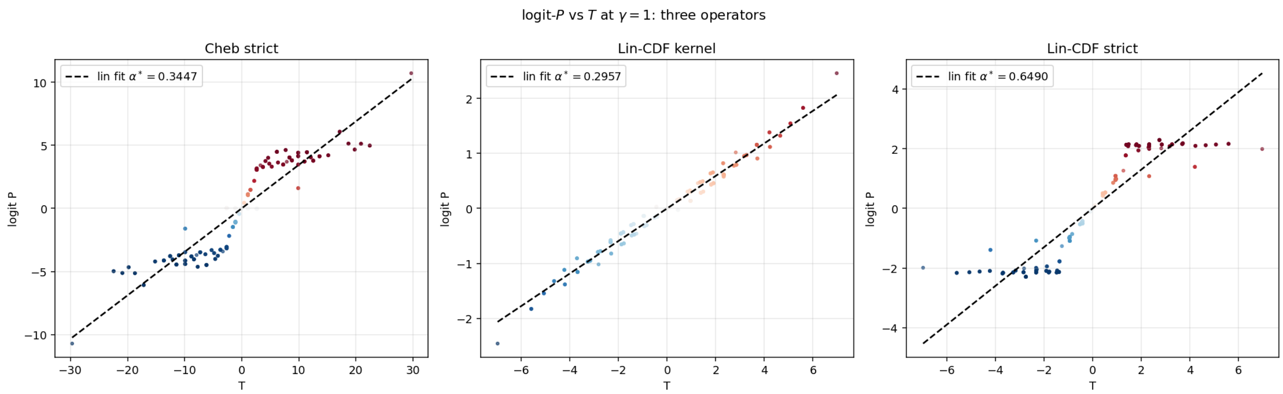


Same equilibrium concept, three different numerical pictures.

4 Lin-CDF strict contour gallery



5 logit-P vs T at $\gamma = 1$



6 Summary

- **For the actual REE** at strict h=0: use Cheb strict POU. Slope $\alpha^* = 0.345$ at γ=1, one-to-one deficit 0.0285.
- **For fast parametric sweeps**: use Lin-CDF kernel band. Fast, robust, all γ reach machine ε — but FP is the regularised version (smaller slope, near-zero one-to-one deficit).
- **Lin-CDF strict h=0 does not work** for high-precision FP at this G — derivative jumps prevent Newton from converging below 10^{-2}.

The Chebyshev representation is *required* for strict-h=0 machine-ε convergence: it provides the C^∞ derivatives that the strict co-area integrand needs.